

# IBM Capstone Project

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Bijaya Basnet  
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# OUTLINE

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- Executive Summary
- Introduction
- Methodology
- Results
  - Visualization – Charts
  - Dashboard
- Discussion
  - Findings & Implications
- Conclusion
- Appendix

# EXECUTIVE SUMMARY

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- The analysis examined 18,845 Stack Overflow survey responses.
- Established technologies dominate current usage:
  - JavaScript is the most-used language.
  - PostgreSQL is the leading database.
  - AWS is the most-used cloud platform.
- Future preferences favor TypeScript, Go, Rust, Redis, and modern frameworks.
- Approximately 68.6% of respondents are between 25 and 44 years old.
- Bachelor's and master's degree holders represent approximately 72.3% of respondents.



# INTRODUCTION

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## Purpose

- Identify the technologies developers use today and want to use next year.

## Audience

- Technology learners, employers, and workforce planners.

## Value

The results highlight durable core skills and emerging technologies that may influence training and hiring priorities.



# METHODOLOGY

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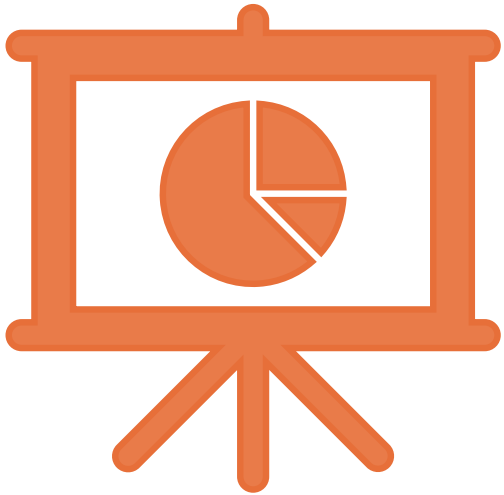


- Source: 2024 Stack Overflow Developer Survey dataset
- • Records analyzed: 18,845 respondents
- • Multi-select technology fields were separated at semicolons
- • Each technology was counted by respondent
- • Top 10 values were ranked for current and desired use
  - Null values were excluded where necessary
  - Dashboards were built in Google Looker Studio



# RESULTS

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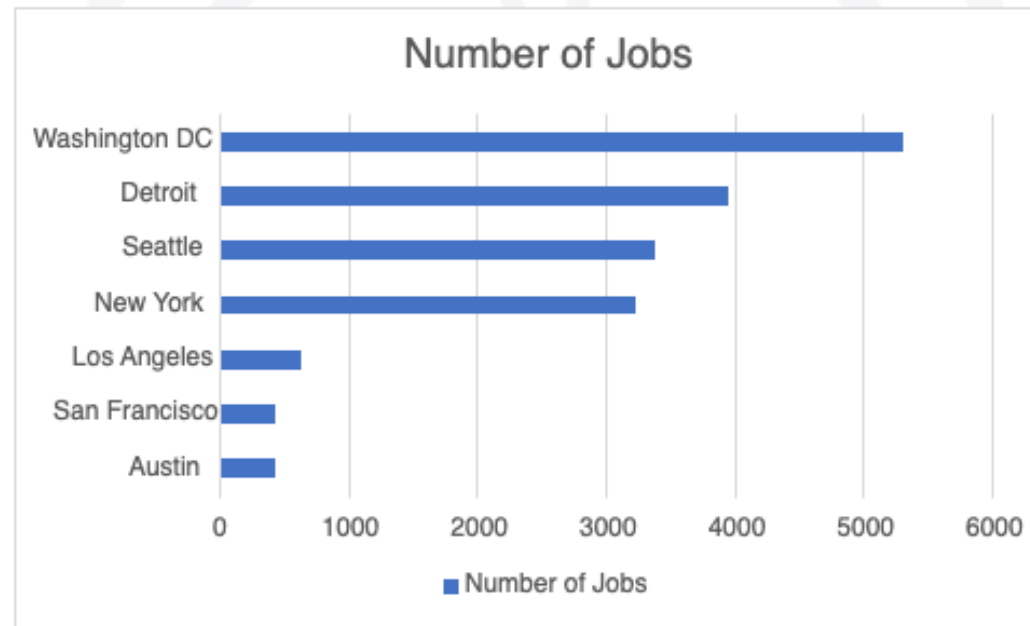
Current usage, future preferences, and demographics reveal a stable core technology stack with selective interest in newer tools.



# JOB POSTINGS

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In Module 1 you have collected the job posting data using Job API in a file named “job-postings.xlsx”. Present that data using a bar chart here. Order the bar chart in the descending order of the number of job postings.



# POPULAR LANGUAGES

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In Module 1 you have collected the job postings data using web scraping in a file named “popular-languages.csv”. Present that data using a bar chart here. Order the bar chart in the descending order of salary.

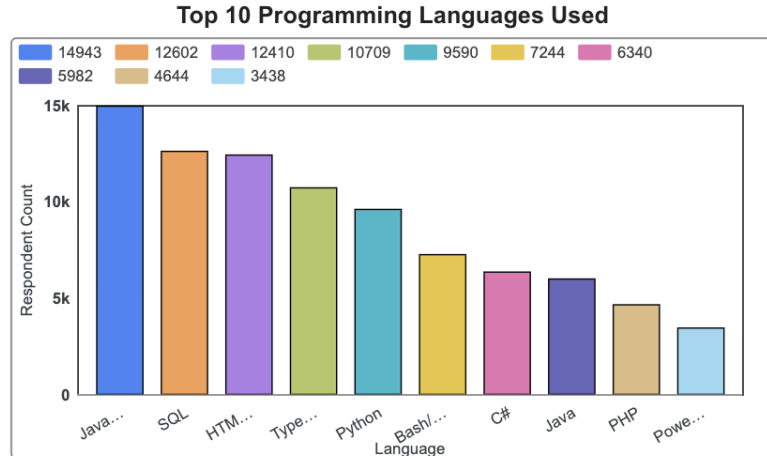




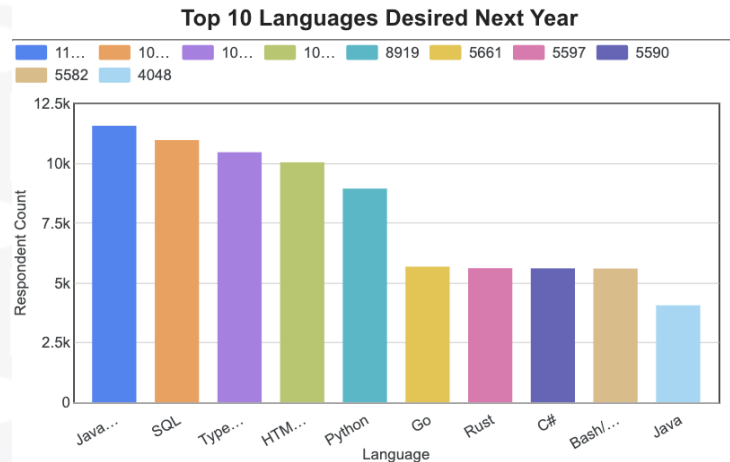
# PROGRAMMING LANGUAGE TRENDS

Summarize key trends shown in the charts.

## Current Year



## Next Year



# PROGRAMMING LANGUAGE TRENDS — FINDINGS & IMPLICATIONS

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## FINDINGS

- • JavaScript remains #1
- • TypeScript moves to #3 in desired use
- • Go and Rust enter the desired top 10
- • Python remains a core general-purpose skill

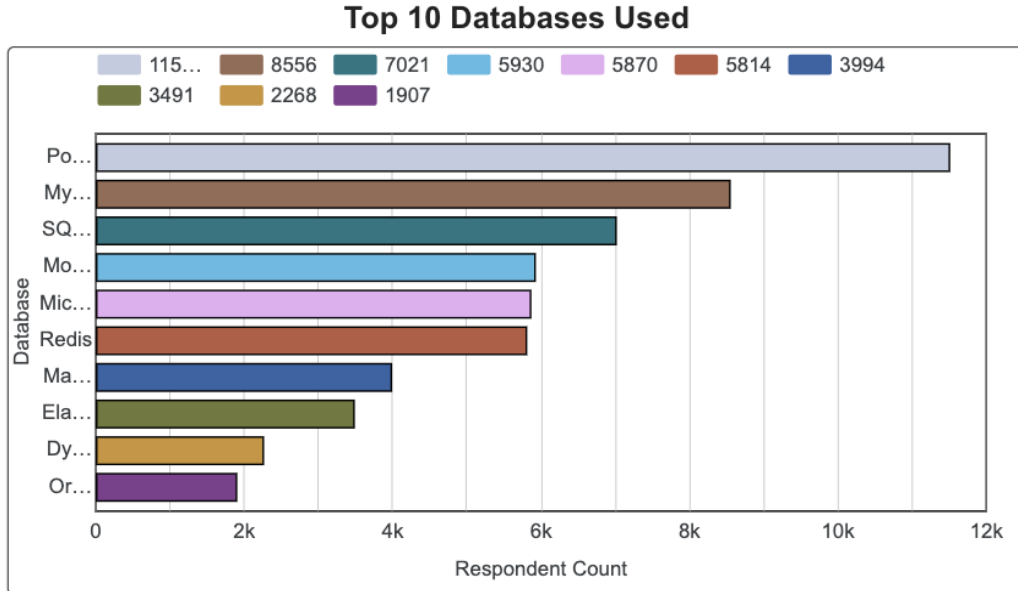
## IMPLICATIONS

- • Keep JavaScript, SQL, and Python in core curricula
- • Pair JavaScript training with TypeScript
- • Add elective pathways for Go and Rust
- • Emphasize transferable programming concepts

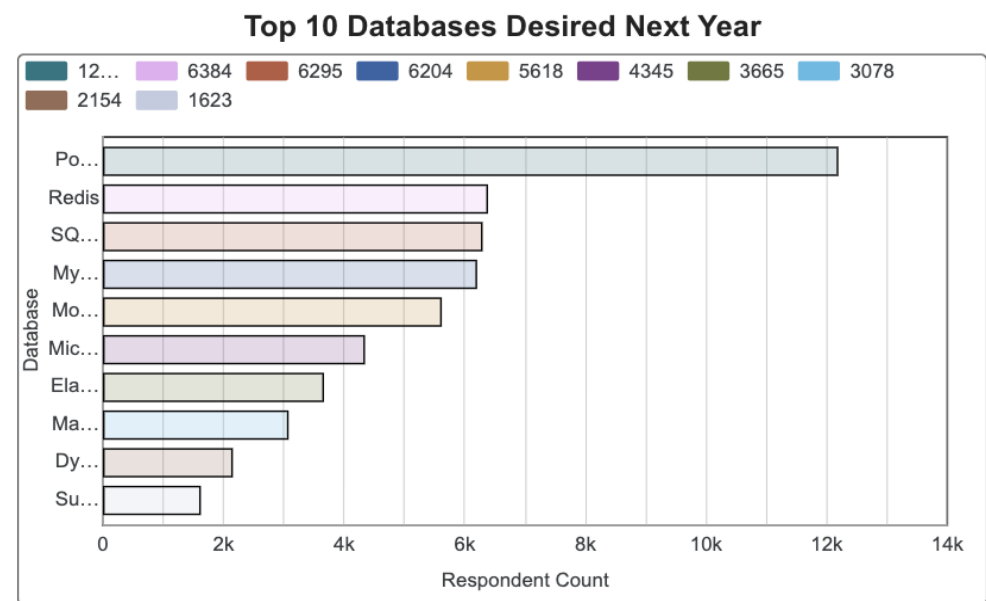
# DATABASE TRENDS

Summarize key trends shown in the charts.

## Current Year



## Next Year



# DATABASE TRENDS — FINDINGS & IMPLICATIONS

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## FINDINGS

- PostgreSQL leads both rankings
- Redis rises sharply in desired use
- SQLite remains highly relevant
- MySQL stays popular but loses relative position

## IMPLICATIONS

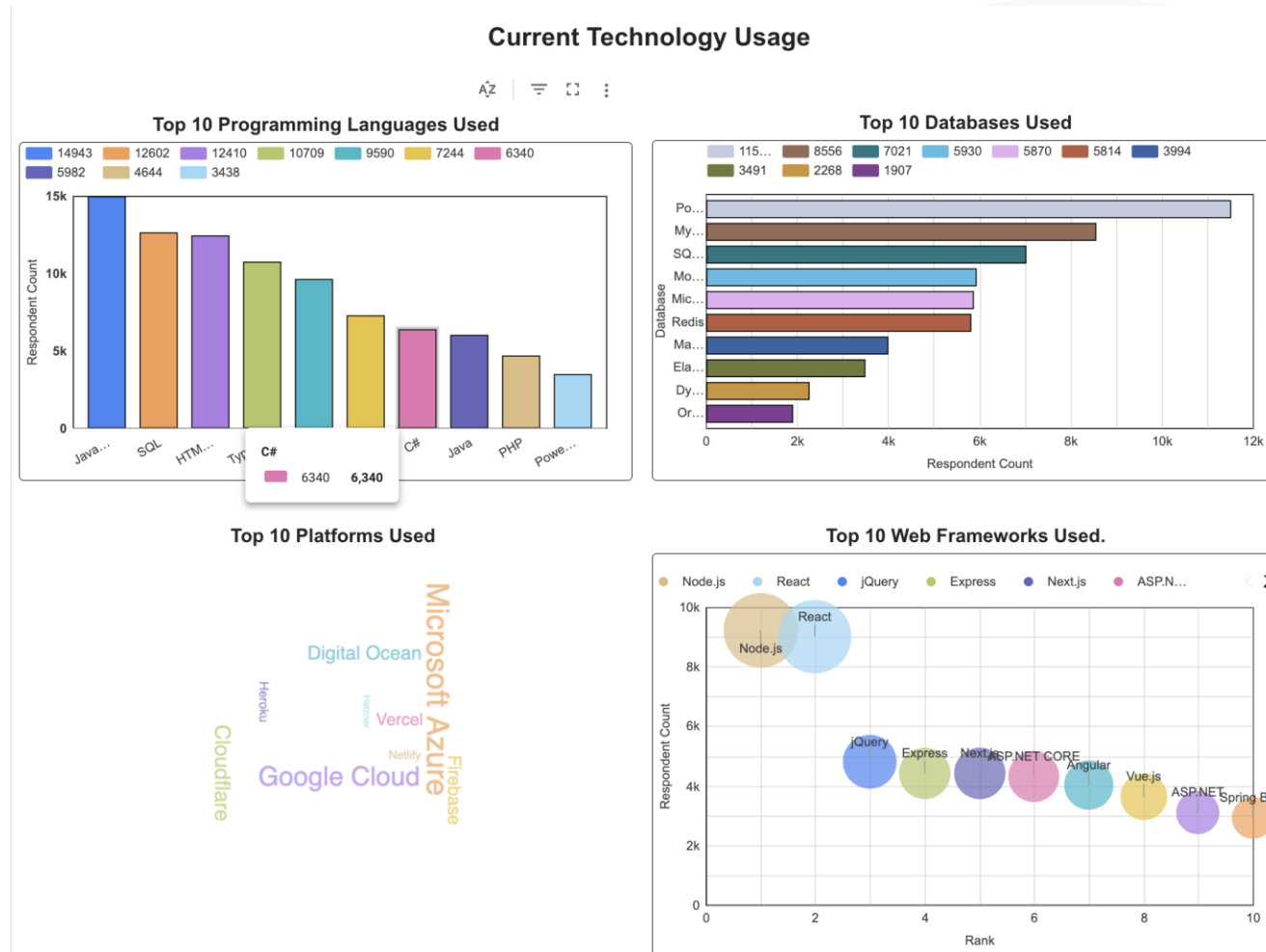
- Prioritize PostgreSQL proficiency
- Teach caching and in-memory data concepts with Redis
- Retain MySQL and SQLite fundamentals
- Cover both relational and NoSQL design choices

# DASHBOARD

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# DASHBOARD TAB 1: Current Technology Usage



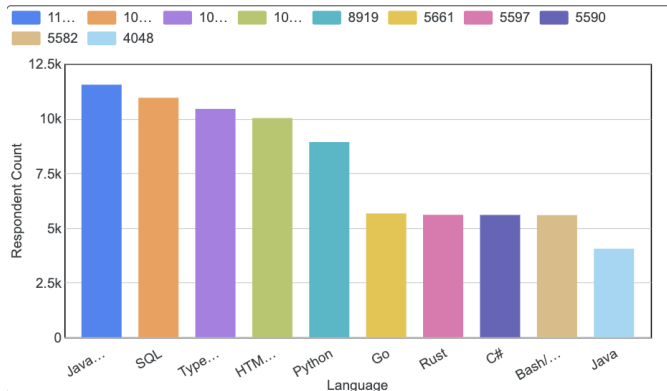
- **Dashboard Insight:** JavaScript, PostgreSQL, AWS, Node.js, and React are the leading technologies currently used by respondents.
- **Key Insights:** Web development, relational databases, and major cloud platforms dominate the current technology landscape.



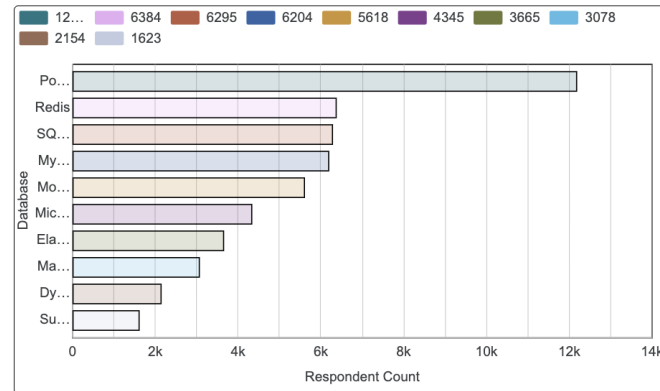
# DASHBOARD TAB 2: Future Technology Trends

## Future Technology Trends

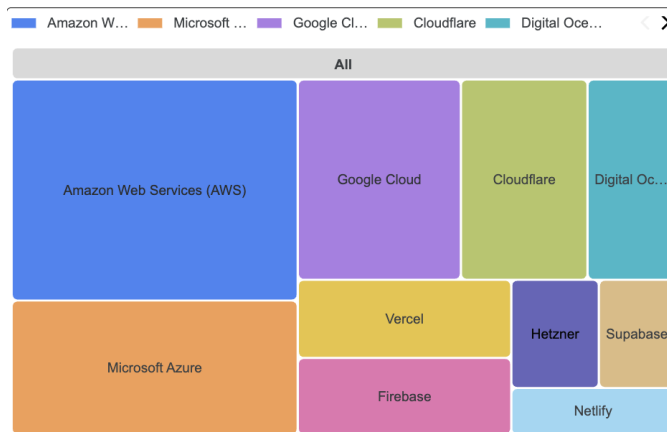
### Top 10 Languages Desired Next Year



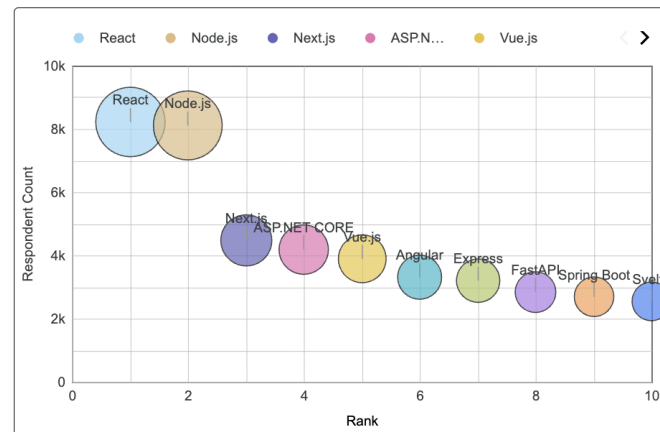
### Top 10 Databases Desired Next Year



### Top 10 Desired Platforms Next Year



### Top 10 Desired Web Frameworks Next Year



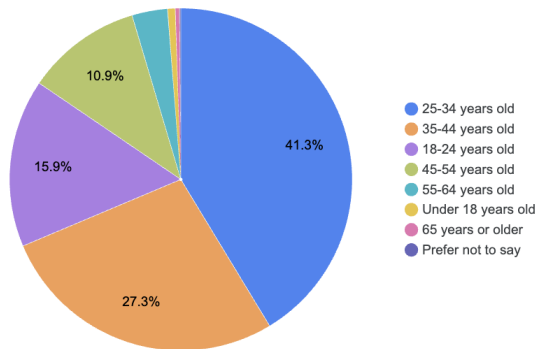
- **Dashboard Insight:** JavaScript, SQL, TypeScript, PostgreSQL, AWS, React, and Node.js remain the most desired technologies.
- **Key Insights:** Developers favor established technologies while showing increased interest in Go, Rust, Redis, Cloudflare, and modern web frameworks.



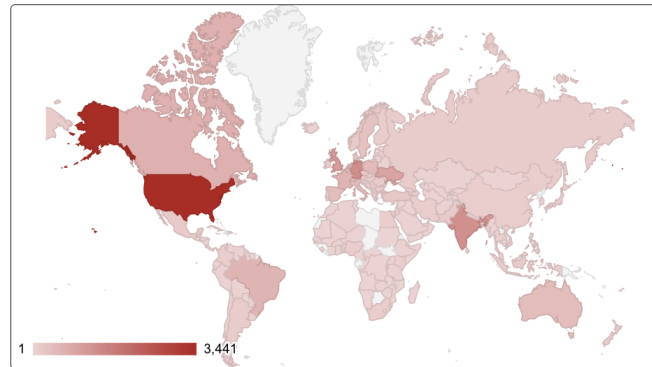
# DASHBOARD TAB 3: Demographics

## Demographics

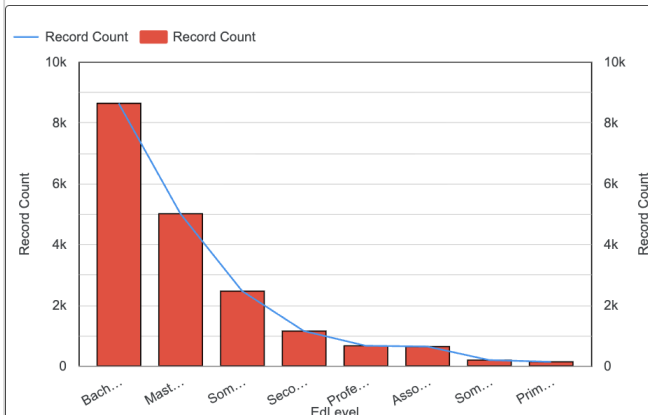
Respondents by Age



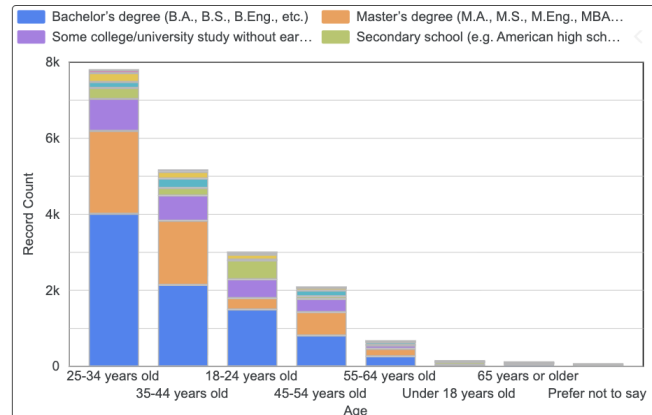
Respondent Count by Country



Respondent Distribution by Education Level



Respondent Count by Age, Classified by Education Level



- **Dashboard Insight:** Most respondents are ages 25–44, live primarily in the United States, and hold bachelor's or master's degrees.
- **Key Insights:** The survey represents a global but highly educated developer population concentrated in the middle stages of their careers.

Data Last Updated: 02/09/2026 19:31:52 | [Privacy Policy](#)





# DISCUSSION

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- The results show continuity rather than wholesale replacement. Developers still rely on established languages and platforms, but desired-use rankings reveal targeted shifts toward typed JavaScript, systems languages, faster data layers, and modern deployment tools. Because the survey permits multiple selections, counts reflect overlapping technology ecosystems.



# OVERALL FINDINGS & IMPLICATIONS

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## FINDINGS

- • Web technologies dominate
- • PostgreSQL is the database leader
- • AWS remains the platform leader
- • Ages 25–44 form 68.6% of respondents

## IMPLICATIONS

- • Build skills around durable foundations
- • Add TypeScript, Go, Rust, and Redis exposure
- • Support cloud-first learning pathways
- • Design training for varied education backgrounds

# CONCLUSION

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- JavaScript, SQL, TypeScript, HTML/CSS, and Python remain essential
- PostgreSQL and AWS anchor the data and cloud ecosystems
- Future interest favors selective modernization—not abandonment of core tools
- The dashboards translate survey responses into practical learning and workforce priorities

# APPENDIX

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- Definitions
  - Current usage: technology selected under “Have Worked With”
  - Future interest: technology selected under “Want to Work With”
  - Respondent count: number of survey records mentioning a technology
  - Multi-select responses were split using semicolon delimiters
  - Top-10 rankings exclude blank values